SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I M.Gokul (192565067) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

level -3

Analytical problem solving

Assessment TOOL 1

Name: M-Gokul
Reg: 192565067
Course: Artificial intelligence
Code: CSA1714.

Breadth - First Search (BFS) and uniform cost search (UCS) on a weighted Graph.

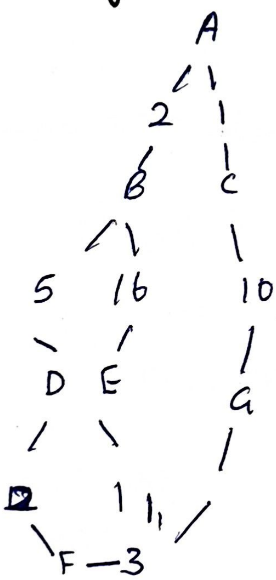
Introduction:

Breadth - first search (BFS) and uniform cost search (UCS) are graph search algorithms used to find paths between nodes.

-> BFS explores nodes level by level and finds the path with the fewest edges.

-> UCS expands the node with the lowest cumulative path cost and finds the least - cost (optimal) path.

Ex weighted graph.



Edge costs.

Edge	Cost
A-B	2
A->C	1
B->D	5
B->E	6
D->F	2
E->G	1
C->G	10
F->G	3

(BFS)

Traversal order.

level 0: A

level 1: B, C

level 2: D, E, G

The first goal node reached is:

A -> C -> G.

Total cost:

A -> C = 1

C -> G = 10

Total cost = 11

uniform Cost Search (UCS)

UCS always expands the node with the smallest
Step-by step.

Step	Expanded Node	Cost
1	A	0
2	B	1
3	C	2
4	D	9
5	F	9
6	E	8
7	G. C (from E)	-9.

Optimal path:

A->B->E->G.

Total Cost:

A->B=2

B->E =6

E->G =1

Total Cost =9.

why BFS is not optimal:

A->C->G

Cost=11

because it has only two edges

UCS chooses:

A->B->E->G

Cost=9.

Conclusion:

Breadth-First Search is suitable only when all edge costs are equal. In weighted graphs with different edge costs, uniform Cost Search is the preferred algorithm because it guarantees the optimal (least-cost) path.

jug ... and 3-Gallon Jug)

Problem Statement

Two jugs are available:

* Jug A = 5 gallons
* Jug B = 3 gallons

Initially, both jugs are empty.

The objective is to measure exactly 4 gallons using only these two jugs and a water pump.

Assumptions.

-> water supply is unlimited

-> Both jugs have no measurement markings.

-> water can be filled completely

-> water can be emptied completely.

State Representation:

(A, B)

A = water in 5-gallon jug

B = water in 3-gallon jug

Initial State

(0, 0)

Goal state.

(4, x) where x can be 0-3 gallons.

Sequence of states.

Step	Action	State (5-gal, 3-gal)
1	fill 5-gallon jug	(5,0)
2	pour into 3-gallon jug	(2,3)
3	Empty 3-gallon jug	(2,0)
4	pour remaining 2 galloons into 3-galloon jug	(0,2)
5	fill 5-galloon jug	(5,2)
6	pour from 5-gallon into 3-gallon.	(4,3).

Goal achieved:

5-gallon jug = 4 galloons.

State diagram:

(0,0)
↓
(5,0)
↓
(2,3)
↓
(2,0)
↓
(0,2)
↓
(5,2)
↓
(4,3) Goal.

Conclusion:

using the operations of filling, emptying and transferring water, exactly 4 gallons can be measured in the 5 gallon jug. this problem demonstrates states - space search in artificial intelligence

Chess playing Agent - PEAS and Task Environment

15-gal, 3-gal,

Introduction

Chess is a classic Artificial Intelligence problem. A chess-playing agent observes the board, analyses possible move, and selects the best move according to the rule of chess.

PEAS = performance measure, Environment, Actuators, sensors

PEAS Description.

performance Measure.

The chess-playing agent should:

* win the game
* Capture opponent's piece
* protect its own king.
* Avoid illegal moves
* Achieve checkmate in minimum moves

Environment.

The environment Consists of:

* chessboard (8 x 8)
* white and Black piece
* chess rule
* opponent.

Actuators:

The agent perform actions by,

* Moving pieces
* Capturing opponent pieces.
* Castling.
* piece promotion.
* [illegal]

Sensor:

The agent receives information through

* Current board position
* opponents move.
* piece locations.
* legal move generator.

Task Environment characteristics:

1. Observable
2. Deterministic
3. Episodic or Sequential
4. Static or Dynamic
5. Discrete or Continuous
6. Single Agent or Multi-Agent

Conclusion:

A chess-playing agent operates in a fully observable, deterministic, sequential, static, discrete, and multi, agent environment. These characteristics make an ideal domain for AI algorithms such as Minimax, Alpha-Beta pruning, and heuristic evaluation functions.